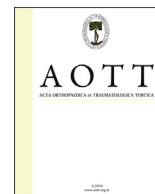




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Pain level after ACL reconstruction: A comparative study between free quadriceps tendon and hamstring tendons autografts

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ABSTRACT

Objective: The objective of this study was to compare the pain levels and analgesic consumption after single bundle ACL reconstruction with free quadriceps tendon autograft versus hamstring tendon autograft.**Patients and methods:** A total of 48 patients scheduled for anatomic single-bundle ACL reconstruction were randomized into two groups: the free quadriceps tendon autograft group (24 patients) and the hamstring tendons autograft group (24 patients). A basic multimodal analgesic postoperative program was used for all patients and rescue analgesia was provided with tramadol, at pain scores over 30 on the Visual Analog Scale. The time to the first rescue analgesic, the number of doses of tramadol and pain scores were recorded. The results within the same group were compared with the Wilcoxon signed test. **Results:** Supplementary analgesic drug administration proved significantly higher in the group of subjects with hamstring grafts, with a median (interquartile range) of 1 (1.3) dose, compared to the group of subjects treated with a quadriceps graft, median = 0.5 (0.1.25) ($p = 0.009$). A significantly higher number of subjects with a quadriceps graft did not require any supplementary analgesic drug (50%) as compared with subjects with hamstring graft (13%; Z-statistics = 3.01, $p = 0.002$). The percentage of subjects who required a supplementary analgesic drug was 38% higher in the HT group compared with the FQT group. **Conclusion:** The use of the free quadriceps tendon autograft for ACL reconstruction leads to less pain and analgesic consumption in the immediate postoperative period compared with the use of hamstrings autograft.**Level of Evidence:** Level I Therapeutic study© 2017 Turkish Association of Orthopaedics and Traumatology. Publishing services by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Introduction

Primary anterior cruciate ligament (ACL) reconstruction is a well-known and researched topic due to its social and economic implications. One of the aspects that still raises controversy is the type of graft used for reconstruction. Lately, the use of the quadriceps tendon as an autograft option is increasing because it has proved to be a versatile graft with low morbidity at the harvest site.^{1,2} Still, the most commonly used autografts for ACL reconstruction are the bone-patellar tendon-bone (BPTB) and hamstring tendons (HT). There is some evidence to suggest that

BPTB autografts produce more pain than HT autografts both in the immediate postoperative period as well as in the long term.³ A 10-year follow-up study by Pinczewski et al compared the pain outcomes in patients that received either a BPTB or HT autograft.⁴ Harvest site symptoms such as tenderness, irritation, and numbness were significantly more common in the BPTB patients.^{3,4} HT autograft is usually recognized to produce the least donor site morbidity, especially compared with the BPTB.³ As compared to HT and BTB autografts, literature data is limited in terms of the free quadriceps tendon (FQT) graft morbidity and pain following its use for ACL reconstruction in the acute postoperative phase.

The objective of this study was to compare the pain levels and analgesic consumption after single bundle ACL reconstruction with a FQT autograft versus a HT autograft in the immediate postoperative phase.

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Patients and methods

Patients and randomization

We designed a prospective longitudinal randomized parallel clinical trial, which took place between October 2013 and May 2015 in our department. The study was approved by our University's ethics committee (298/28.07.2014). After signing informed consent for participation in the study 48 patients scheduled for ACL reconstruction were included.

The inclusion criteria were: patients between 16 and 50 years with a documented ACL tear scheduled for arthroscopic reconstruction, willing to participate in the study.

Exclusion criteria were: patients with associated meniscus tears requiring suture repair, associated ligamentous lesions that required surgical management, previous surgeries on the same knee, patients with chronic pain, hepatic impairment, alcoholics, drug abusers, any analgesic use within 14 days of admission as well as those who refused to participate in the study.

The patients were admitted 24 h before the surgery and no analgesics were administered in this period. The 48 patients were randomized in one of the two groups in the morning of the surgery using free randomization software from Sealed Envelope⁵ to arthroscopic ACL reconstruction with FQT (n = 24) or HT (n = 24).

Surgical technique

All surgeries were performed by the same surgeon (AT) and the same surgical technique was used except for the graft harvest. We performed arthroscopic anatomic single-bundle ACL reconstruction, placing the tunnels in the center of the footprints both on the femur and on the tibia, using a three portal technique.^{6,7} We used extracortical button femoral fixation and bioabsorbable interference screw tibial fixation in every case. The FQT graft was harvested through a 4 cm median longitudinal incision, using a No. 10 blade to establish the depth and a clean cut at the apex of the patella to release the tendon.⁸ The HT graft was harvested in the classical manner through a 3 cm antero-medial longitudinal incision taking care not to damage the infrapatellar branch of the saphenous nerve.⁹

All patients received spinal anesthesia and no other peripheral nerve block was associated. Spinal anesthesia was induced in lateral or sitting position with 15–20 mg 0.5% spinal plain bupivacaine, at the L2–L3 intervertebral space, using pencil-point 25–27 G spinal needles.

Postoperative pain evaluation and management

A multimodal analgesic postoperative regimen was started 1 h after completion of surgery and consisted of acetaminophen (Paracetamol, Terapia-Ranbaxy, Cluj-Napoca, Romania) 500 mg per os (po) and ketorolac tromethamina (Ketorol, Dr. Reddy's Lab. (UK) LTD.) 15 mg intravenous (iv) every 8 h, for the first 48 postoperative hours. Rescue analgesia was provided with tramadol (Aliud[®] Pharma GmbH & Co. KG, Gottlieb-Daimler-Str. 19, D-89150 Leichingen, Germany) 30 mg iv bolus when patients declared pain scores of 30 or more on the Visual Analog Scale (VAS).

Postoperative pain intensity was evaluated using the Visual Analog Scale, ranging from 0 to 100, with 0–30 considered mild pain, 30 to 70 moderate pain and over 70, severe pain.

Time to the first rescue analgesic requirement, the number of doses of tramadol and the pain score were recorded. The postoperative period was split three-way (first 12 h postop, 13–24 h postop, 25–48 h postop). In the first 24 h the pain was recorded with the patients in bed-rest and, in the third period, in-between exercises.

All patients underwent the same hospital stay regime and were discharged on day 3 after surgery. On day 1 after surgery the wounds were inspected and drain removed. All local complications were noted (swelling, redness, numbness or any other local symptoms). All patients were encouraged to ambulate with axillary support with toe touch weight bearing for bathroom privileges only on day one after surgery and as tolerated from day 2 on. All the patients were advised to keep the knee in full extension throughout the day, except for 15 min of passive flexion exercises 3 times a day.

All procedures performed were in accordance with the ethical standards of the institutional and/or National Research Committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Statistical analysis

Parametric data were expressed as mean $\pm$ standard deviation (SD) and categorical data as median [interquartile range (IR)].

The median values and interquartile range are presented for each group. The results within the same group were compared with Wilcoxon signed test. Differences less than 0.05 were considered statistically significant when two groups were compared. Statistical analysis was performed by using R version 2.15.1 software¹⁰ (R Core Team, Vienna, Austria).

Results

48 subjects were included in the study and half of them were treated with a FQT graft and the other half with a HT graft. The majority of subjects included in the study were male patients (M:F = 3.8), the difference between the percentage of male and female being statistically significant ($p < 0.0001$). The mean age of subjects included in the study was 28.35 ± 7.19 years. No significant difference was registered between genders regarding the age (M: 28.82 ± 6.35 years; F: 26.60 ± 9.98 years; $p = 0.391$) or between treatment groups (FQT: 29.21 ± 8.52 years; HT: 27.50 ± 5.62 years; $p = 0.416$). Also, no significant difference was registered regarding the body mass index (BMI) between treatment groups (FQT: 26.27 ± 4.85 ; HT: 25.68 ± 3.32 ; $p = 0.6232$).

The median number of the supplementary analgesic drug administration (tramadol, 30 mg/dose) proved significantly higher in the group of subjects with HT, with a median [IR] of 1^{1.3} dose, compared to the group of subjects treated with FQT median 0.5 [0.1,25] dose (Z-statistics = -2.61, p-value = 0.009). A significantly higher number of subjects with FQT did not receive any rescue analgesic (50%) compared with patients with HT (13%; Z-statistics = 3.01, p-value = 0.002).

The detailed results of patients who required supplementary analgesia are presented in Table 1.

The percentage of subjects who required supplementary analgesic drugs was 38% higher in the group with HT compared with the group with FQT (Fig. 1).

The pain scores on the visual scale for the patients who required supplementary analgesics were similar between groups (Table 2).

Table 1

Results separated between the postoperative periods with the patients who required supplementary analgesia.

Postoperative interval	FQT (n = 24)		HT (n = 24)	
	n	%	N	%
0–12 h	12	50.00 (29.34; 70.66)	18	75.00 (54.34; 91.49)
13–24 h	5	20.83 (08.51; 41.49)	12	50.00 (29.34; 70.66)
25–48 h	4	16.66 (04.34; 37.33)	9	37.50 (16.84; 58.16)

n = number of patients who required supplementary analgesia.

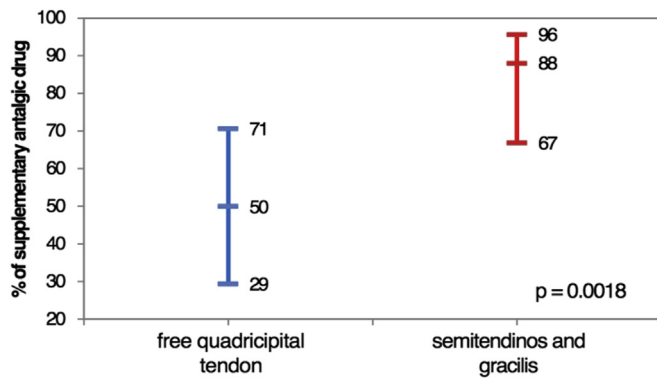


Fig. 1. Percentage and associated 95% confidence interval for receiving supplementary analgesic drug by groups.

Table 2

VAS pain values for patients who requested supplementary analgesia separated between the postoperative periods.

Postoperative interval	FQ (n = 24)		STG (n = 24)	
	n	VAS median (Q1, Q3)	n	VAS median (Q1, Q3)
0–12 h	12	60 (35, 80)	18	50 (40, 62.5)
13–24 h	5	60 (40, 70)	12	47.5 (37.5, 66.25)
25–48 h	4	40 (32.5, 42.5)	9	50 (40, 60)

n = number of patients who required supplementary analgesia.

The mean time from the completion of surgery to the first supplementary analgesic requirement was 8.25 h in the FQT group and 7.85 h in the HT group.

No local complications were noted in any of the two groups during the hospital stay, other than usual tenderness after knee arthroscopy.

There were 3 patients who reported nausea, 2 in the HT group and one in the FQT group. One patient from the HT group that reported nausea also experienced vomiting. These symptoms were interpreted as side effects of opioid treatment and responded well to corresponding treatment.

Discussion

The most important finding of this study is that the use of the FQT autograft for SB ACL reconstruction produces less pain than the HT autograft, in the immediate postoperative period, with statistical significance. In our patient group, this resulted in over 30% less drug consumption in the FQT group than in the HT group. Half of the patients in the FQT group did not request any tramadol, their pain level being minor (under 30 mm – VAS). On the other hand, patients who did request supplementary analgesics reported similar pain scores on the visual scale, between groups. Also, the mean time from the completion of surgery to the first requirement of supplementary analgesic drug was similar between the two groups.

The two groups were similar concerning age and gender distribution, and the male/female ratio was similar to other studies.¹¹

The less invasive harvest technique for FQT probably helped in obtaining these results, since we had no local complication (significant lesion of the quadriceps pouch, unwanted damage to the quadriceps tendon) and we were able to obtain an average of 8.5 cm of graft length with a 4 cm incision, as other authors did.^{12,13}

We chose to compare these two types of autografts because they are both soft tissue without any bone component. Also, we are limiting the use of BPTB grafts due to the fact that studies have shown that using BPTB graft results in higher local morbidity than

HT^{14,15} and more frequent kneeling pain than the quadriceps tendon with bone block.¹⁶ There is data suggesting that any type of graft is suitable for ACL reconstruction on an outpatient basis with regard to early postoperative pain and morbidity,¹⁷ however no data exists with regard to a FQT graft related morbidity in the immediate postoperative phase. Studies showed that multimodal pain therapy following ACL reconstruction is able to control pain levels and allows one-day surgery protocol for this procedure.^{18–21} However, minimizing pain and opioid consumption is very important for the well being, comfort and early recovery of the patient following ACL reconstructive surgery.²² According to our study, using FQT as a graft choice alone reduces drug consumption and limits the need for additional anesthesia or drug mix, and of course their related secondary effects (nausea, vomiting and excessive sedation).

The postoperative analgesic regime used in this study was developed by the anesthesiologist (AHO) and was tested to be optimal before starting the study.

We are aware that there are some limitations of our study. The pain is difficult to measure and the same level of pain may be reported different by patients and is dependent by many factors. We excluded patients with history of analgesic use within 14 days prior to surgery or history of chronic medication that might influence the study. Also, some bias may exist because the surgeon who performed the local clinical evaluations was aware of the surgical procedure. However, the patient reported pain level recording and rescue analgesia was performed by a study nurse which was blinded to the study protocol.

Even though sample size was low, and no power analysis was performed before the study, a statistical significance was obtained and we believe that the difference observed between the two groups regarding pain level and drug consumption would be respected in larger groups.

In conclusion, the use of the FQT autograft for ACL reconstruction leads to less pain and analgesic consumption in the immediate postoperative period compared with the use of hamstrings autograft.

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